From Image to Insight: An Explainable Multimodal Vision-Language Framework for Chest Radiograph Analysis and Triage in Low-Resource Healthcare

A Mini Project Report Submitted in partial fulfillment of the requirements for the award of the degree of

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In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

By

P. Kavya Sai – 2420030322 (Team Lead)

P. Meghna – 2420030401

S. Anjana – 2420030499

D. Poojitha -- 2420030697

Under the Esteemed Guidance of

Dr. K Swanthana



Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

The mini-project report entitled "From Image to Insight: An Explainable Multimodal Vision-Language Framework for Chest Radiograph Analysis and Triage in Low-Resource Healthcare" is a record of bona fide work carried out by P.KavyaSai (2420030322) ,P. Meghna (2420030401) ,S.Anjana (2420030499) and D. Poojitha(2420030697) in partial fulfillment of the requirements for the award of the B.Tech degree at K L (Deemed to be) University. The work presented in this report is original to the best of our knowledge, and all datasets, tools, code, and external sources used have been appropriately acknowledged.

P. Kavya Sai – 2420030322

P. Meghna – 2420030401

S. Anjana – 2420030499

D. Poojitha - 2420030697

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the mini-project report entitled "From Image to Insight: An Explainable Multimodal Vision-Language Framework for Chest Radiograph Analysis and Triage in Low-Resource Healthcare" is a bona fide work carried out by P. KavyaSai (2420030322), P. Meghna (2420030401), S. Anjana(2420030499) and D. Poojitha(2420030697) under the supervision of Dr. K Swanthana during the academic year 2026-2027,in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Signature of the Project Supervisor

Signature of the HOD **Signature of the External Examiner**

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P.Kavya Sai – 2420030322

P.Meghna – 2420030401

S.Anjana – 2420030499

D. Poojitha - 2420030697

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1. ABSTRACT

ABSTRACT

Chest radiograph interpretation is challenging in rural and low-resource healthcare settings where access to radiologists may be limited. This project proposes an explainable multimodal vision-language framework that integrates abnormality classification, visual explanation, semantic cross-verification, clinical report retrieval, and preliminary triage. The framework uses the NIH ChestX-ray14 dataset, containing 112,120 frontal-view chest radiographs across 14 abnormality categories, for classification, and the Indiana University Chest X-Ray (IU X-Ray) dataset from the National Library of Medicine's OpenI collection for clinical report retrieval. Image preprocessing includes resizing, normalization, and preparation for model inference.

A Vision Transformer (ViT-B/16) performs multi-label classification using binary cross-entropy with positive-class weighting to address class imbalance. Zero-shot CLIP provides independent vision-language cross-verification, while BLIP v1 generates supplementary image descriptions. Attention-based heatmaps support visual interpretability, and TF-IDF with cosine similarity retrieves relevant radiologist-authored reports. A deterministic, rule-based triage module assigns preliminary urgency categories. On a held-out test set of 25,596 chest radiographs covering 15 labels, the classifier achieved macro-averaged precision of 0.295, recall of 0.392, and F1-score of 0.327. These results indicate varying performance across conditions and highlight the importance of class-sensitive evaluation. The framework aims to support preliminary screening, reporting, and case prioritization while preserving clinician oversight and avoiding autonomous diagnosis.

Team Name: MedVision AI

Team Members and Responsibilities:

1. P. Kavya Sai- 2420030322 (Role : Explainability and Vision-Language Models)
2. P. Meghna- 2420030401 (Role: Image Preprocessing and ViT Classification)
3. S. Anjana – 2420030499 (Role: Retrieval, Triage, and Report Generation)
4. D.Poojitha – 2420030697 (Role: UI Development, Integration, and Deployment)

1. Dataset/source and project repository : ChestX-ray14, IU X-Ray (Indiana University), and NLM OpenI datasets; GitHub project repository
2. Dataset URL / repository URL
   - ChestX-ray14: https://nihcc.app.box.com/v/ChestXray-Nihcc
   - IU X-Ray / OpenI: https://openi.nlm.nih.gov/
   - GitHub repository: https://github.com/Kavya20060702/KLH-CSE-2026-2027-2420030322-Explainable-Vision-Language-Diagnostic-Assistant-for-Chest-Radiographs.git
   - Hugging Face model repository: https://huggingface.co/kavyasaibp/chest-xray-classifier

2. INTRODUCTION

Chest radiography is one of the most widely used imaging techniques for identifying thoracic abnormalities and supporting medical assessment. However, in rural and low-resource healthcare settings, the limited availability of radiologists can delay image interpretation and subsequent clinical decisions. Manually examining large numbers of chest radiographs requires specialized expertise, considerable time, and careful attention to subtle visual patterns. These challenges create a need for computer-assisted systems that can provide preliminary analysis and help healthcare professionals prioritize cases efficiently.

Machine learning and deep learning techniques can identify complex patterns in medical images and assist in detecting potential abnormalities. However, a conventional image classification model generally produces disease labels and confidence scores without providing sufficient visual evidence, relevant clinical context, or urgency recommendations. To address these limitations, this project proposes an explainable multimodal vision-language framework that combines image classification, vision-language analysis, visual explanation, clinical report retrieval, and preliminary triage within a unified workflow.

The intended users are healthcare professionals, particularly those working in resource-constrained settings. The primary input is a chest radiograph, and the system produces predicted thoracic abnormalities with confidence scores, attention-based visual explanations, supplementary image descriptions, vision-language cross-verification, relevant retrieved radiology reports, and a preliminary urgency category. These outputs are intended to support professional review rather than replace clinical judgment.

The proposed pipeline begins with data collection using the NIH ChestX-ray14 dataset for abnormality classification and the Indiana University Chest X-Ray (IU X-Ray) dataset from the National Library of Medicine's OpenI collection for clinical report retrieval. Images undergo preprocessing, including resizing and normalization. A Vision Transformer (ViT-B/16) performs multi-label classification, while zero-shot CLIP provides semantic cross-verification and BLIP v1 generates supplementary image descriptions. Attention-based heatmaps provide visual explanations, and TF-IDF with cosine similarity retrieves relevant radiologist-authored reports. A deterministic, rule-based module then assigns a preliminary triage category. The classifier is evaluated using precision, recall, F1-score, and other class-sensitive metrics to account for class imbalance. A Streamlit-based interface presents the analysis results in a structured format.

The remainder of this paper is organized as follows: Section II discusses the problem statement and research objectives; Section III reviews related work and identifies the research gap; subsequent sections describe the proposed system, datasets, preprocessing, model architecture, implementation, and evaluation methodology. The final sections present experimental results, discuss limitations, and conclude with future directions.

3. LITERATURE SURVEY

3.1 Literature Survey

1. ChestX-ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases (Wang et al., 2017)

- Problem and dataset: The study addressed large-scale chest X-ray analysis using 108,948 frontal-view images from 32,717 patients, labeled for eight thoracic diseases.
- Method: It developed a weakly supervised, multi-label classification and disease-localization framework using labels extracted from radiology reports.
- Evaluation and findings: Experiments demonstrated that disease classification and localization were feasible using image-level labels without requiring detailed annotations for every image.
- Limitation: Report-derived labels can be noisy, and the framework does not provide the complete combination of independent vision-language verification, clinical report retrieval, and triage.
- Relevance: This work establishes the dataset-based foundation for automated chest radiograph analysis. [1]

2. CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning (Rajpurkar et al., 2017)

- Problem and dataset: The study investigated automated pneumonia detection using the NIH ChestX-ray14 dataset, containing more than 100,000 frontal-view chest X-rays.
- Method: CheXNet used a 121-layer DenseNet model for pneumonia detection and extended the approach to all 14 disease categories.
- Evaluation and findings: The authors compared model performance against four practicing academic radiologists on a specially annotated test set and reported higher average F1-score performance for pneumonia detection.
- Limitation: The primary contribution is disease classification and localization; it does not integrate the full clinical text retrieval and rule-based triage workflow proposed here.
- Relevance: It provides a prominent deep-learning baseline for chest X-ray classification. [2]

3. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization (Selvaraju et al., 2017)

- Problem and dataset: The researchers addressed the difficulty of understanding which image regions influence deep neural network predictions, evaluating visual explanations across image classification and other vision tasks.
- Method: Gradient-weighted Class Activation Mapping (Grad-CAM) uses gradients flowing into a CNN's final convolutional layer to produce localization heatmaps.
- Evaluation and findings: The authors evaluated localization performance and conducted human studies, reporting that the explanations helped users distinguish stronger from weaker models even when their predictions were identical.
- Limitation: Conventional Grad-CAM relies on suitable convolutional feature maps and does not itself provide clinical language, disease severity, or triage decisions.
- Relevance: This work motivates visual interpretability. The present project instead uses Vision Transformer attention maps, not conventional gradient-based Grad-CAM. [3]

4. An Image Is Worth 16×16 Words: Transformers for Image Recognition at Scale

(Dosovitskiy et al., 2021)

- Problem and dataset: The work investigated whether image recognition could use a pure Transformer architecture instead of relying primarily on convolutional neural networks.
- Method: Vision Transformer (ViT) divides images into fixed-size patches, embeds them as tokens, and uses self-attention to learn relationships between image regions.
- Evaluation and findings: The model was evaluated on image-recognition benchmarks, including ImageNet, CIFAR-100, and VTAB, achieving competitive results after large-scale pretraining.
- Limitation: Strong performance depends on pretraining and suitable computational resources; general image-recognition results do not guarantee equivalent performance in medical imaging.
- Relevance: This research provides the architectural foundation for the ViT-B/16 classifier used in the proposed framework. [4]

5. Learning Transferable Visual Models From Natural Language Supervision (Radford et al., 2021)

- Problem and dataset: The study addressed the limited flexibility of image classifiers trained on fixed categories by learning from image–text pairs collected from the internet.
- Method: CLIP uses contrastive learning to align image and text representations, allowing text prompts to specify candidate categories at inference time.
- Evaluation and findings: The authors evaluated CLIP across more than 30 vision datasets and demonstrated zero-shot transfer to diverse recognition tasks.
- Limitation: General-purpose pretraining does not guarantee reliable interpretation of medical radiographs, and zero-shot similarity scores are not equivalent to clinically validated diagnoses.
- Relevance: CLIP motivates the independent image–text cross-verification component of the proposed system. [5]

6. BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation (Li et al., 2022)

- Problem and dataset: This research addressed limitations of vision-language models that perform well either at understanding images or generating text, rather than both.
- Method: BLIP combines vision-language understanding and generation with caption bootstrapping, using synthetic captions and filtering to improve noisy image–text training data.
- Evaluation and findings: The authors reported improvements in image–text retrieval, image captioning, and visual question answering across established benchmarks.
- Limitation: The original model was developed using general image–text data, so generated captions are not necessarily medically accurate or suitable as clinical findings.
- Relevance: The project uses BLIP v1 to provide supplementary image descriptions, not independent diagnostic conclusions. [6]

7. ConVIRT: Contrastive Learning of Medical Visual Representations from Paired Images and Text (Zhang et al., 2022)

- Problem and dataset: The authors addressed the shortage and expense of expert-labelled medical images by exploiting naturally occurring paired medical images and descriptive

reports.

- Method: ConVIRT applies a bidirectional contrastive objective to learn visual representations from paired images and text.
- Evaluation and findings: The method was evaluated on four medical image classification tasks and two zero-shot retrieval tasks. The authors reported better or comparable classification performance using approximately 10% of the labelled data required by an ImageNet-initialized counterpart.
- Limitation: Its training approach depends on paired medical image–text data and focuses on representation learning rather than a complete explainability-to-triage workflow.
- Relevance: It demonstrates the value of clinical text for medical image understanding and supports the multimodal motivation of this project. [7]

8. GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-Efficient Medical Image Recognition (Huang et al., 2021)

- Problem and dataset: GLoRIA addresses the challenge of training medical image models when large quantities of manually labelled data are difficult to obtain.
- Method: It learns global image–text representations and local relationships between image regions and words in paired radiology reports through attention-based contrastive learning.
- Evaluation and findings: Experiments covered image–text retrieval, supervised and zero-shot classification, and segmentation, demonstrating label-efficient medical image representations.
- Limitation: The framework concentrates on multimodal representation learning and downstream recognition tasks; it does not provide the same integrated, rule-based urgency assessment and structured clinical-support workflow.
- Relevance: GLoRIA demonstrates why linking image regions to clinical language can provide richer evidence than image-only classification. [8]

9. MedCLIP: Contrastive Learning from Unpaired Medical Images and Text (Wang et al., 2022)

- Problem and dataset: The researchers addressed the limited availability of paired medical image–report datasets and the false negatives that can arise during conventional contrastive learning.
- Method: MedCLIP decouples images and text during multimodal contrastive learning and uses medical knowledge-based semantic matching.
- Evaluation and findings: The authors reported improvements over comparison methods in zero-shot prediction, supervised classification, and image–text retrieval, including experiments with substantially less pretraining data.
- Limitation: Its main contribution is medical vision-language representation learning, not a complete workflow combining visual explanations, retrieved reports, and deterministic triage.
- Relevance: MedCLIP provides an important medical-domain comparison for the project's CLIP-based cross-verification approach. [9]

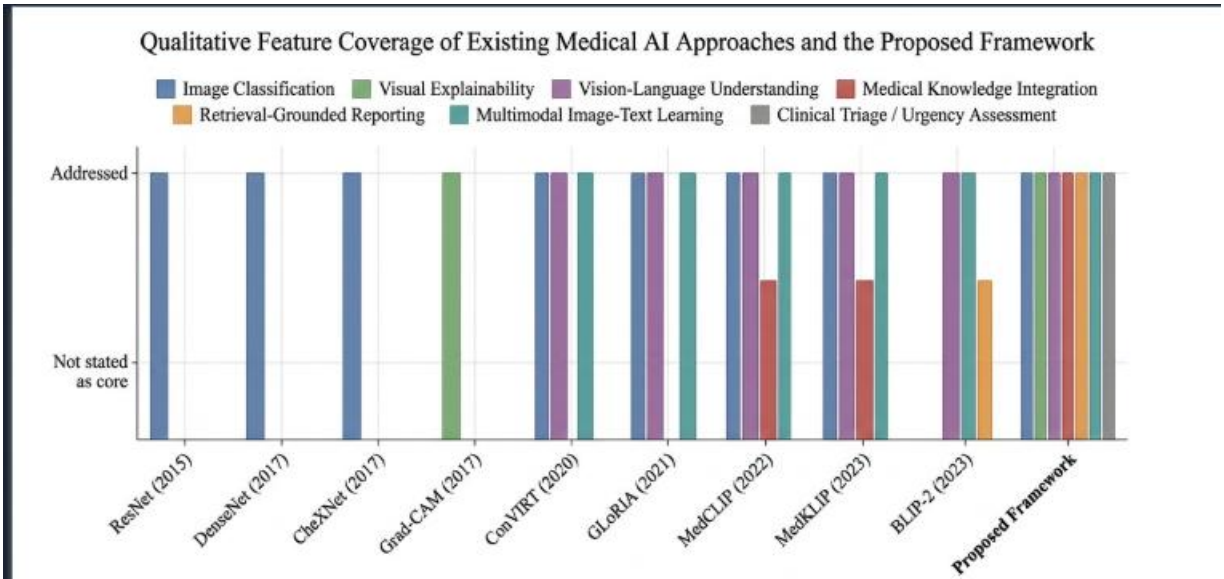


Figure X. Qualitative comparison of functional feature coverage across representative medical AI approaches and the proposed explainable multimodal framework.

3.3 Research gap and contribution of the proposed mini-project

The literature review identifies four main gaps.

Research Gap Analysis Table

S. No.	Existing Research / Approach	Research Gap / Limitation	Proposed Solution in Our Framework
1	ChestX-ray8 (Wang et al., 2017)	Relies on report-derived labels that may contain noise and does not integrate the complete explainability, verification, and triage workflow.	Use a ViT-based multi-label classifier combined with complementary evidence and a structured analysis pipeline.
2	CheXNet (Rajpurkar et al., 2017)	Primarily focuses on pneumonia detection and chest X-ray classification rather than an integrated clinical-support workflow.	Combine multi-abnormality classification with vision-language verification, report retrieval, and rule-based triage.
3	Grad-CAM (Selvaraju et al., 2017)	Provides visual localization but does not independently provide clinical language, severity assessment, or triage decisions.	Generate Vision Transformer attention-based heatmaps to support interpretation of model predictions.
4	Vision Transformer (ViT) (Dosovitskiy et al., 2021)	General image-recognition performance does not guarantee reliable medical-image interpretation; pretraining and computational resources can be important.	Adapt ViT-B/16 for chest radiograph classification and evaluate it on a held-out test set.
5	CLIP (Radford et al., 2021)	General-purpose image-text alignment may not reliably represent medical findings; similarity scores are not	Use image-text similarity as a supplementary cross-check alongside the primary classifier.

		clinically validated diagnoses.	
6	BLIP (Li et al., 2022)	Generated descriptions may be medically inaccurate because the original model was trained primarily on general image-text data.	Use BLIP-generated descriptions only as supplementary evidence, not as independent diagnostic conclusions.
7	ConVIRT (Zhang et al., 2022)	Focuses on learning medical visual representations from paired images and reports rather than the complete explanation-to-triage workflow.	Integrate vision-language evidence into a broader preliminary assessment pipeline.
8	GLoRIA (Huang et al., 2021)	Emphasizes global and local image-text representation learning but does not provide the proposed deterministic triage and structured reporting workflow.	Combine visual explanations and image-text evidence with severity assessment and structured report generation.
9	MedCLIP (Wang et al., 2022)	Addresses medical image-text representation learning but does not by itself integrate report retrieval, visual explanations, and urgency recommendations.	Incorporate vision-language cross-verification as one component of a unified framework.
10	Conventional image-only classifiers	Typically produce disease predictions without integrating retrieved clinical context, independent text-based evidence, and auditable urgency rules.	Combine classification, explainability, retrieval-grounded reporting, and deterministic triage in one system.

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

To develop an explainable multimodal machine-learning framework that analyzes chest radiographs, identifies potential thoracic abnormalities, retrieves relevant radiologist-authored reports, and provides preliminary urgency recommendations to support healthcare professionals in low-resource settings.

Problem Description

In rural and low-resource healthcare settings, limited access to radiologists can delay chest X-ray interpretation and preliminary assessment. Manual examination of large numbers of radiographs requires specialized expertise and time, while conventional classification models may provide predictions without sufficient visual explanations or clinical context. This project integrates image classification, vision-language cross-verification, attention-based visualization, TF-IDF-based clinical report retrieval, and rule-based triage to support preliminary assessment. The system is intended to assist healthcare professionals and does not replace clinical diagnosis or medical judgment.

Objectives

1. To obtain, inspect, and document the NIH ChestX-ray14 dataset for abnormality classification and the Indiana University Chest X-Ray (IU X-Ray) dataset from the NLM OpenI collection for clinical report retrieval, including their sources, sizes, labels, and relevant data fields.
2. To preprocess chest radiographs through resizing, normalization, and input-format standardization before model inference.
3. To implement a Vision Transformer (ViT-B/16) classifier for multi-label prediction of thoracic abnormalities using binary cross-entropy loss and positive-class weighting to address class imbalance.
4. To incorporate zero-shot CLIP for independent image–text cross-verification and BLIP v1 for supplementary image descriptions.
5. To generate attention-based heatmaps that visually represent image regions associated with the classifier's predictions.
6. To implement TF-IDF vectorization and cosine similarity for retrieving relevant radiologist-authored reports from the OpenI clinical report corpus.
7. To develop a deterministic, rule-based triage module that assigns preliminary urgency categories: Routine, Soon, Urgent, and Immediate.
8. To evaluate the primary classifier using precision, recall, F1-score, AUROC, and class-wise performance analysis on a held-out test set.
9. To integrate the implemented components into a Streamlit-based interface that presents predictions, confidence scores, visual explanations, retrieved clinical context, and preliminary triage recommendations.
10. To analyze classification errors, class imbalance, model limitations, potential biases, privacy considerations, and future improvements while maintaining human oversight throughout the workflow.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

- **Processor:** AMD Ryzen 5 5500U processor, with 6 CPU cores and 12 logical processors, as specified in the experimental setup.
- **RAM:** 13.8 GB available system memory in the reported development environment.
- **GPU:** Two NVIDIA Tesla T4 GPUs with 16 GB memory each, providing a total of 32 GB GPU memory for accelerated model training and inference.
- **Storage:** NVMe SSD for storing datasets, pretrained models, source code, checkpoints, and experimental results.
- **Operating System:** Windows 11 Home Single Language, as specified in the experimental setup.
- **Internet Connectivity:** Required for downloading datasets, pretrained models, and software dependencies. Offline operation may be possible after the necessary resources are downloaded and configured.
- **Dataset Storage:** Sufficient storage for the NIH ChestX-ray14 images, IU X-Ray images and reports, model weights, and generated outputs.

Software Requirements

- **Programming Language:** Python 3.x.
- **Development Environment:** Kaggle Notebook for cloud-based model development and experimentation.
- **Deep Learning Framework:** PyTorch for implementing and training the Vision Transformer classifier.
- **Model Libraries:** Hugging Face Transformers for pretrained vision-language models and related model components.
- **Data Processing:** NumPy and pandas for numerical operations and data handling.
- **Image Processing:** OpenCV and appropriate image-transformation utilities for resizing, normalization, and image preparation.
- **Machine Learning and Evaluation:** scikit-learn for TF-IDF vectorization, cosine similarity utilities where applicable, and classification evaluation metrics.
- **Vision-Language Models:** ViT-B/16 for primary classification, CLIP for image–text cross-verification, and BLIP v1 for supplementary image descriptions.
- **Clinical Text Retrieval:** TF-IDF vectorization and cosine similarity for retrieving relevant radiology reports from the OpenI corpus.
- **User Interface:** Streamlit for presenting predictions, visual explanations, retrieved clinical context, and preliminary triage results.
- **Datasets:** NIH ChestX-ray14 for chest radiograph classification and the Indiana University Chest X-Ray (IU X-Ray) dataset from the NLM OpenI collection for radiology-report retrieval.
- **Version Control:** Git, optionally with GitHub, for source-code management and project collaboration.

6. IMPLEMENTATION

6 .1 Data and Task

The project implements a multimodal framework for chest radiograph analysis, abnormality classification, and preliminary triage. It combines image-based prediction with clinical text retrieval to provide supporting information for healthcare professionals.

Dataset and source

1. NIH ChestX-ray14 dataset

The ChestX-ray14 dataset contains 112,120 frontal-view chest radiographs from the NIH Clinical Center and is used for training and evaluating the primary Vision Transformer (ViT-B/16) classifier. It contains 14 thoracic abnormality categories. The implementation evaluates 15 labels, including No Finding and the 14 abnormality categories.

2. Indiana University Chest X-Ray (IU X-Ray) dataset

The IU X-Ray dataset contains chest radiographs paired with radiology reports. It is used for image–text analysis and retrieval of relevant clinical reports. The associated NLM OpenI collection provides access to biomedical images and their accompanying text.

The datasets were obtained from their public distributions through Kaggle and OpenI. The complete datasets are not included in the project repository because of their size and source distribution conditions. Users must comply with the applicable access terms and licence or usage restrictions of each dataset**.**

Task definition

The primary task is multi-label image classification, in which a chest radiograph can be associated with multiple abnormalities. The secondary text-analysis task retrieves relevant radiologist-authored reports from the clinical report corpus using TF-IDF and cosine similarity.

The classifier was evaluated on a held-out test set of 25,596 radiographs. The paper states that patient separation is maintained across training, validation, and test data. Validation data are used for model selection and class-specific threshold tuning. Exact training and validation sample counts are not reported in the paper.

6.2 NLP Preprocessing and Feature Extraction

The clinical text retrieval module uses TF-IDF (Term Frequency–Inverse Document Frequency) to convert radiology reports into numerical feature vectors. This representation gives greater weight to terms that are informative within a report and reduces the relative importance of terms that occur frequently across the corpus.

The retrieval process consists of the following steps:

1. **Clinical report corpus**: Radiologist-authored reports from the IU X-Ray/OpenI collection serve as the text source.
2. **Query construction:** A query is created from the available analysis and predicted findings.
3. **TF-IDF representation:** The query and candidate reports are represented numerically.
4. **Cosine similarity:** Similarity scores are calculated between the query and candidate report vectors.
5. **Report ranking:** Reports are ranked by similarity, and a relevant report is selected to provide supporting clinical context.

The retrieved report is incorporated into the structured preliminary assessment. It is treated as reference information rather than a verified interpretation of the current radiograph.

The research paper does not specify the exact handling of missing report text, duplicate removal, lowercasing, tokenization, stop-word removal, stemming, or lemmatization. These preprocessing details should be documented from the actual implementation rather than assumed.

6.3 Model Training and Evaluation

Primary classifier

The primary classifier is an ImageNet-pretrained Vision Transformer (ViT-B/16) with a task-specific linear classification head. The model processes a radiograph as a sequence of image patches and generates a prediction for each target condition.

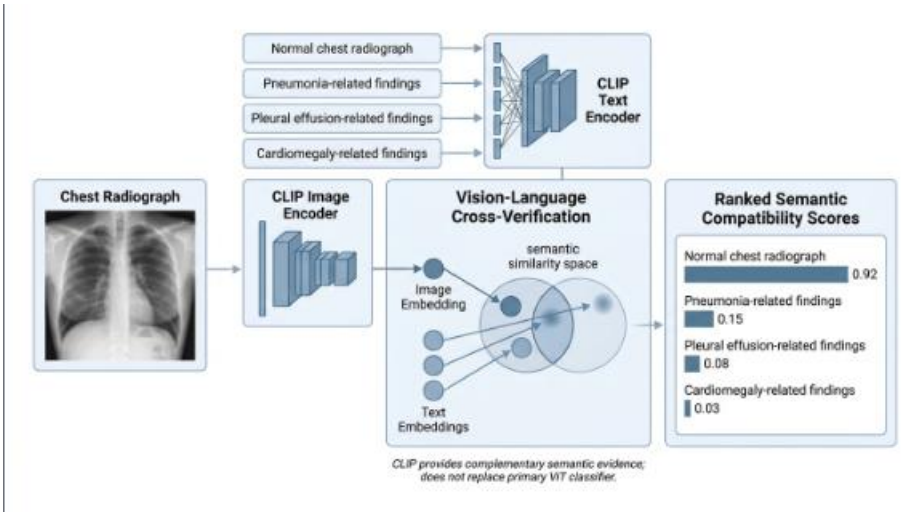
Training uses binary cross-entropy with logits and positive-class weighting to address class imbalance. The classifier produces independent probabilities for the 15 target labels.

The best checkpoint is selected according to validation mean AUROC. After checkpoint selection, a separate decision threshold is chosen for each class using the validation precision–recall curve, selecting the threshold that maximizes F1-score. These thresholds are then applied to the held-out test predictions.

Supporting components

- **CLIP:** A pretrained, zero-shot vision-language model compares radiographs with text descriptions of candidate abnormalities to provide an independent semantic cross-check.
- **BLIP v1:** Generates supplementary image descriptions. These descriptions are not treated as definitive clinical findings.
- **Attention-based visualization**: Extracts the Vision Transformer's attention from the classification process and produces heatmaps over image patches.
- **TF-IDF retrieval:** Represents clinical reports numerically and ranks them using cosine similarity.
- **Rule-based triage:** Uses deterministic rules to assign preliminary urgency categories: Routine, Soon, Urgent, and Immediate.

  The framework integrates these modules through application-level orchestration. It does not train a new end-to-end multimodal fusion network.



Evaluation metrics

The classifier is evaluated using accuracy, precision, recall, F1-score, and AUROC. Per-class and macro-averaged results are considered because the dataset has substantial class imbalance.

The reported results on the held-out test set are:

Metric	Reported result
Macro-averaged accuracy	0.866
Macro-averaged precision	0.295
Macro-averaged recall	0.392

Macro-averaged F1-score	0.327

The results show that classification performance varies considerably across conditions. Consequently, accuracy alone is not sufficient to assess the model's ability to identify abnormalities, especially those with low prevalence.

The paper does not report the complete set of numerical hyperparameters, random seed, or exact sample counts for the training and validation splits. These details should be added only if they can be verified from the code or experiment logs.

6.4 Project Workflow and Modules

The project is organized into the following functional modules.

1. Data loading and exploratory analysis

Loads the chest radiograph data, labels, and associated reports. The data are inspected to understand their structure and the distribution of abnormality labels.

2. Image preprocessing

Standardizes radiograph inputs through resizing, normalization, and preparation for the pretrained models.

3. Clinical text retrieval

Processes the report corpus using TF-IDF, calculates cosine similarity against the query, and retrieves relevant radiologist-authored reports.

4. Feature extraction and classification

Uses ViT-B/16 for multi-label abnormality prediction and confidence estimation. CLIP provides an independent image–text cross-check, while BLIP v1 generates supplementary image descriptions.

5. Explainability

Uses the Vision Transformer's attention over image patches to generate a heatmap associated with the primary prediction. This provides visual evidence for human review but does not establish that the highlighted region is pathological.

6. Evaluation and error analysis

Evaluates the primary classifier on the held-out test set using class-wise and aggregate metrics. The analysis considers class imbalance and differences in performance across conditions.

7. Structured reporting and triage

Combines predicted abnormalities, confidence scores, attention-based evidence, vision-language outputs, retrieved clinical context, and preliminary urgency information into a structured decision-support report.

The paper describes these functional modules but does not specify the exact source filename for each module. The actual Python filenames should be added from the project repository.

6.5 Machine-Learning Implementation

The framework uses specialized components for different tasks rather than a collection of conventional text classifiers.

The ViT-B/16 model serves as the primary predictive model. It is trained using binary cross-entropy with logits and positive-class weighting. CLIP and BLIP v1 provide complementary vision-language information, while TF-IDF and cosine similarity perform clinical report retrieval. The triage component applies deterministic rules to the available findings and confidence information.

No Multinomial Naive Bayes, Logistic Regression, Linear SVM, Decision Tree, or Random Forest comparison is reported as part of the implemented primary classification pipeline. These algorithms should not be listed as trained comparison models unless they are subsequently implemented and evaluated.

Key implementation code should be referenced in the appendix or project repository. The source files should cover data loading, image preprocessing, ViT training and inference, TF-IDF report

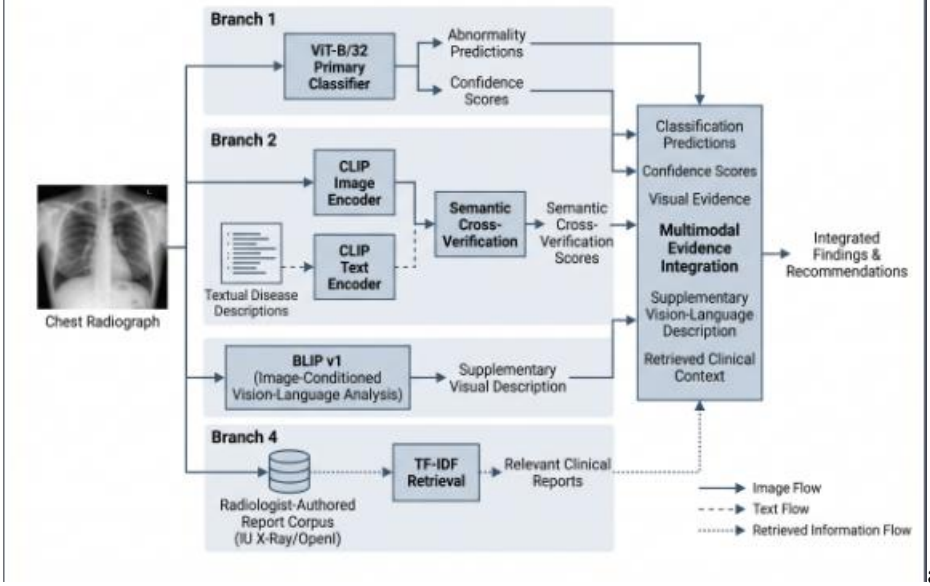
retrieval, attention visualization, evaluation, and triage.

6.6 Demonstration Interface

The framework includes a Streamlit-based interface for presenting the analysis results. The interface brings together the primary classification predictions, confidence scores, attention-based visual explanations, supplementary vision-language outputs, retrieved clinical reports, and preliminary triage recommendations.

The interface provides a structured view of the evidence generated by the different modules, allowing a healthcare professional to review the outputs alongside the original radiograph. The retrieved clinical report is presented as supporting context, and the urgency category is treated as a preliminary prioritization recommendation. Final interpretation and clinical decisions remain the responsibility of qualified healthcare professionals.

Figure: **Multimodal Evidence Integration Architecture for Chest Radiograph Analysis**



analysis

7. RESULTS AND DISCUSSION

7.1 Dataset Summary

The proposed framework was evaluated using the NIH ChestX-ray14 dataset and the IU X-Ray dataset available through the OpenI collection. ChestX-ray14 contains 112,120 frontal-view chest radiographs covering 14 thoracic disease categories. The classification task includes 15 output labels: *No Finding* and 14 abnormality categories. A held-out test set containing 25,596 samples was used to evaluate the classification model. The IU X-Ray dataset, which contains chest radiographs paired with radiology reports, was used to support image–text analysis and clinical report retrieval.

The dataset was divided to maintain separation between training, validation, and testing data. The validation set was used for model selection and threshold optimization, while the held-out test set was used to report the final evaluation results.

Table 1. Dataset Summary

Dataset	Purpose	Description
NIH ChestX-ray14	Image classification	112,120 frontal-view chest radiographs covering 14 thoracic disease categories
IU X-Ray (OpenI)	Image–text analysis and report retrieval	Chest radiographs paired with radiologist-authored reports
Held-out test set	Final evaluation	25,596 samples

7.2 Preprocessing and Class Distribution

The chest radiographs were standardized before being passed to the Vision Transformer (ViT-B/16) classification model. The preprocessing stage included image resizing, normalization, standardization, and image-quality checking. These steps helped provide consistent image inputs for model inference.

The dataset contains multiple disease categories, and the number of samples is not evenly distributed across all conditions. This class imbalance can affect classification performance, particularly for less frequently represented abnormalities. To address this issue during training, binary cross-entropy with logits and positive-class weighting were used. The final model thresholds were selected using validation precision–recall curves by maximizing the F1-score for each class.

The text-retrieval component used TF-IDF features and cosine similarity to identify reports relevant to the analysis. The retrieved reports provide supporting clinical context and are not treated as verified reports for the current patient.

7.3 Model Implementation and Comparison

The framework combines several pretrained and task-specific components. ViT-B/16 serves as the primary multi-label image classifier. CLIP provides an independent image–text comparison using textual prompts, while BLIP generates supplementary image descriptions. Attention-based heatmaps help visualize image regions that influence the model, and TF-IDF with cosine similarity supports the retrieval of relevant radiology reports. A rule-based triage module assigns preliminary urgency categories: Routine, Soon, Urgent, and Immediate.

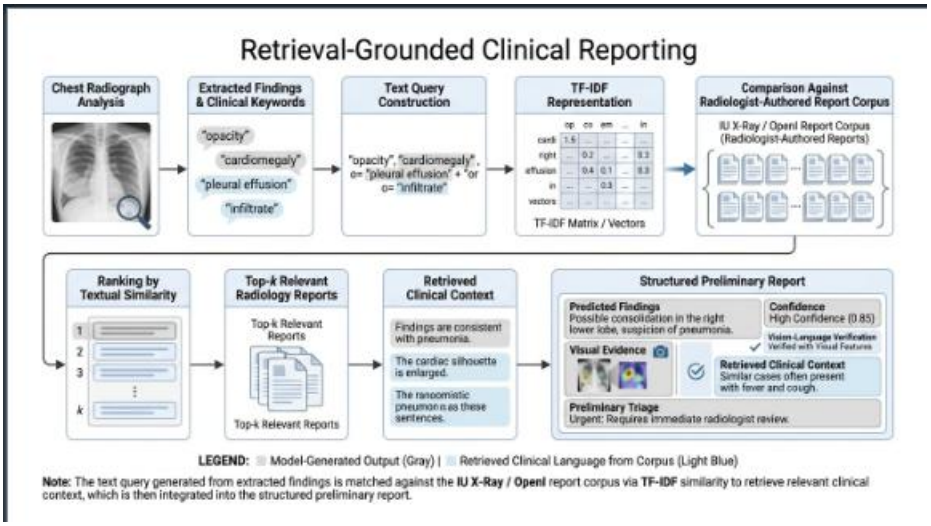


Table 2. Model Components and Their Roles

Component	Method used	Purpose
Primary classifier	ViT-B/16	Multi-label classification of chest radiographs
Training objective	Binary cross-entropy with logits and positive-class weighting	Train the classifier while accounting for class imbalance
Semantic cross-check	Zero-shot CLIP	Compare images with text prompts describing candidate conditions
Image description	BLIP v1	Generate supplementary image descriptions
Visual explanation	Transformer attention-based heatmap	Highlight image patches receiving attention
Clinical report retrieval	TF-IDF and cosine similarity	Retrieve relevant radiology reports
Triage	Rule-based decision logic	Assign preliminary urgency categories

The components are coordinated at the application level. The framework does not introduce a newly trained end-to-end multimodal architecture or a learned fusion layer. Each component contributes a specific function to the overall analysis workflow

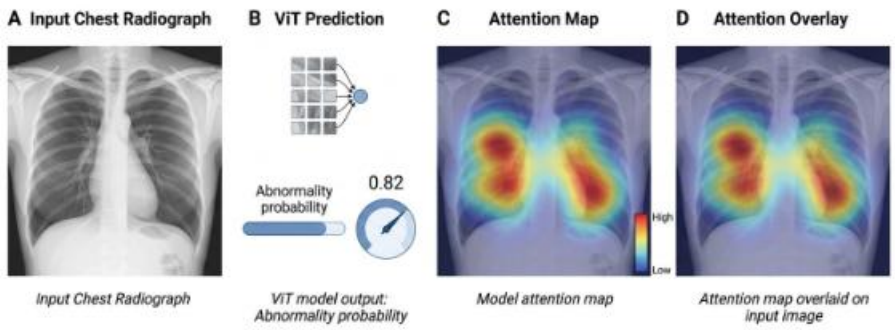


Figure Legend: Four-panel illustration of explainable AI applied to synthetic chest radiograph analysis. **(A)** Input image. **(B)** Vision Transformer (ViT) model output: predicted abnormality probability (no diagnostic claim). **(C)** Model attention map highlights regions influencing prediction. **(D)** Attention map transparently overlaid to localize model focus on anatomical context.

7.4 Evaluation Results

The primary classification model was evaluated using accuracy, precision, recall, F1-score, and AUROC. The best model checkpoint was selected using validation mean AUROC, and class-specific thresholds were determined using validation precision–recall curves. The reported held-out test results are summarized below.

Table 3. Overall Test Performance

Evaluation metric	Reported result
Macro-averaged precision	0.295
Macro-averaged recall	0.392
Macro-averaged F1-score	0.327
Macro-averaged accuracy	0.866

The results show that the model achieved a macro-averaged accuracy of 0.866, while macro-averaged precision, recall, and F1-score were lower. This difference indicates that accuracy alone does not fully describe the model's ability to identify individual disease categories. The precision, recall, and F1-score results therefore provide important additional information about the model's classification performance.

Table 4. Selected Class-Wise Evaluation Results

Disease category	Reported metric	Result
Effusion	F1-score	0.508
Infiltration	Recall	0.739
Pneumonia	Precision	0.066
Pneumonia	F1-score	0.086
Hernia	Accuracy	0.994
Hernia	F1-score	0.108

The class-wise results demonstrate that performance varies considerably between disease categories. Effusion achieved an F1-score of 0.508, while Infiltration achieved a recall of 0.739. In contrast, Pneumonia showed low precision and F1-score, and Hernia had high accuracy but a low F1-score. These results illustrate why class-wise precision, recall, and F1-score are necessary when evaluating

a multi-label medical image classifier.

7.5 Discussion of Errors and Limitations

The evaluation results indicate that the framework has potential as a decision-support system, but its performance is not uniform across all disease categories. The low precision and F1-score for Pneumonia suggest difficulty in producing reliable positive predictions for that category. Similarly, the low F1-score for Hernia, despite its high accuracy, shows that accuracy can be misleading when a category is imbalanced or positive examples are relatively uncommon.

Positive-class weighting and validation-based threshold selection were incorporated to address class imbalance during training and prediction. However, these techniques do not eliminate the effects of imbalanced data or guarantee strong performance for every condition. The results also highlight the need for further evaluation on diverse clinical datasets and careful assessment of false-positive and false-negative predictions.

The explainability component uses attention-based heatmaps derived from the final transformer block. These visualizations indicate which image patches receive attention, but they do not prove that a highlighted region contains a disease. Likewise, CLIP comparisons, BLIP descriptions, and retrieved radiology reports provide supplementary context rather than independent confirmation of a diagnosis. The rule-based triage output should therefore be interpreted as preliminary prioritization, not as a substitute for professional clinical assessment.

7.6 Summary

The experimental evaluation demonstrates that the proposed framework integrates chest radiograph classification, image–text comparison, image description, attention-based visualization, clinical report retrieval, and rule-based triage in a single application workflow. The primary classifier achieved a macro-averaged precision of 0.295, recall of 0.392, F1-score of 0.327, and accuracy of 0.866 on the reported test set. The differences in class-wise performance and the limitations associated with imbalanced data indicate that further validation is necessary before practical clinical use. The framework should be considered a research-oriented decision-support tool rather than an autonomous diagnostic system.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project addresses the challenge of analyzing chest radiographs in healthcare settings where access to timely radiologist assessment may be limited. To address this problem, an explainable multimodal vision-language framework was developed to combine image classification, visual explanation, image–text comparison, clinical report retrieval, and preliminary triage within a single application.

The framework uses a Vision Transformer (ViT-B/16) as the primary multi-label classification model for identifying 14 thoracic abnormalities and the *No Finding* category. Additional components include zero-shot CLIP for semantic cross-verification, BLIP v1 for supplementary image descriptions, attention-based heatmaps for visual interpretation, TF-IDF with cosine similarity for retrieving relevant radiology reports, and a rule-based module for assigning triage categories. These components are coordinated at the application level to provide structured analysis and supporting clinical context.

The experimental evaluation on the held-out test set of 25,596 samples achieved a macro-averaged precision of 0.295, recall of 0.392, F1-score of 0.327, and accuracy of 0.866. The class-wise results showed variations in performance across disease categories, demonstrating that accuracy alone is insufficient to assess a medical image classification system. Although the framework integrates the intended analysis components, the classification results indicate that further improvement is needed, particularly for categories with low precision and F1-scores.

The project objectives were addressed by implementing the primary classification model, integrating complementary vision-language components, providing attention-based visual explanations, retrieving relevant radiology reports, and developing rule-based triage logic. However, the results do not establish clinical diagnostic reliability, and the framework should be regarded as a decision-support prototype rather than an autonomous diagnostic system.

The most important limitation is the variation in classification performance across disease categories, influenced in part by class imbalance. In addition, attention heatmaps do not confirm the presence of disease, retrieved reports may not directly correspond to the current patient's condition, and the system requires further validation before clinical deployment.

8.2 Future Scope

The following improvements can extend the capabilities and reliability of the proposed framework:

1. **Evaluation on larger and more diverse datasets:** Test the framework using additional chest radiograph datasets from different hospitals, patient populations, and imaging environments to assess generalization.
2. **Improved class-imbalance handling:** Investigate alternative loss functions, sampling strategies, and class-specific threshold optimization to improve the precision, recall, and F1-score of underperforming disease categories.
3. **Advanced vision-language models:** Evaluate improved pretrained vision-language models and compare their performance with the existing ViT-B/16 and CLIP components while maintaining a clear distinction between classification and supporting analysis.
4. **Enhanced explainability:** Assess attention-based heatmaps against expert annotations or other suitable explanation methods to determine whether highlighted image regions are clinically meaningful.
5. **Improved clinical report retrieval:** Explore transformer-based text representations, including BERT, as an alternative or complement to TF-IDF for retrieving reports that

better match the predicted findings and clinical context.

6. **Deployment and latency testing:** Measure inference time, memory consumption, and resource requirements under different hardware configurations. Evaluate the feasibility of deployment in low-resource healthcare environments before claiming real-time or CPU-only operation.
7. **Monitoring and privacy-aware processing:** Investigate model-performance monitoring for changes in incoming data and implement appropriate privacy safeguards for medical images and radiology reports.

   These extensions could improve the framework's generalization, interpretability, and practical usability. Further experimental validation and clinical review would remain necessary before the system could be considered suitable for real-world medical decision support.

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9. REFERENCES

9.1. REFERENCES

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